Last time: Given a power series $\sum c_n x^n$, find the radius of convergence

$$R = \frac{1}{\limsup |c_n|^{1/n}} \in [0,\infty]$$

Then: ① For $x \in (-R,R)$, $\sum c_n x^n$ converges to a continuous function.

② For $|x| > R$, $\sum c_n x^n$ diverges.

Ex $\sum_{n=0}^{\infty} \frac{x^n}{2^n}$

Set $c_n = \frac{1}{2^n}$. Then

$$|c_n|^{\frac{1}{n}} = \frac{1}{2} \quad \forall n \geq 1 \quad \Rightarrow \quad \limsup |c_n|^{\frac{1}{n}} = \frac{1}{2}$$

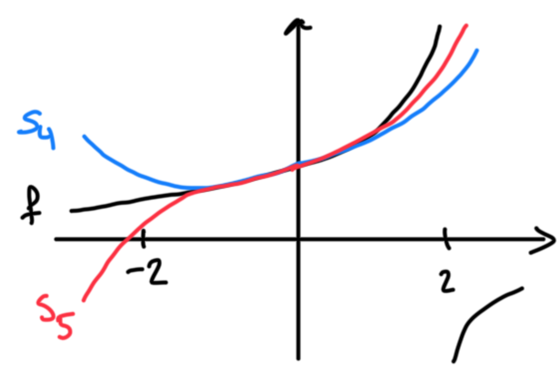
$$\Rightarrow \quad R = \frac{1}{\limsup |c_n|^{\frac{1}{n}}} = 2$$

So $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{2^n}$ ...

- converges for each $x \in (-2,2)$
- converges uniformly for $x \in [-2+\varepsilon, 2-\varepsilon]$, $\forall \varepsilon > 0$
- diverges for $|x| > 2$.

Note that this is a geometric series:

$$f(x) = \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n = \frac{1}{1-\frac{x}{2}}$$



This series...

- Does not converge at $x = \pm 2$:

$$\sum \left(\frac{\pm 2}{2}\right)^n = \sum (\pm 1)^n \quad \text{diverges}$$

- Does not converge uniformly on $(-2,2)$:

$$d_{sup}(s_n, s_{n-1}) = \sup_{x \in (-2,2)} \left| \sum_{k=0}^{n} \frac{x^k}{2^k} - \sum_{k=0}^{n-1} \frac{x^k}{2^k} \right|$$

$$= \sup_{x \in (-2,2)} \left| \frac{x^n}{2^n} \right| = 1$$

$\Rightarrow$ $\{s_n\}$ not Cauchy in $C((-2,2))$.

TAYLOR SERIES (Rudin Ch. 5)

Def Let $I \subseteq \mathbb{R}$ be an interval, $f: I \to \mathbb{R}$, and $a \in I$. We say:

1. f is differentiable at a if

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} \quad \text{exists.}$$

When this is true, we call the limit the derivative of f at a, and we denote it by $f'(a)$ or $\frac{df}{dx}(a)$.

2. f is differentiable if it is differentiable at a $\forall a \in I$.

Def Let $I \subseteq \mathbb{R}$ be an open interval, $f: I \to \mathbb{R}$, and $x_0 \in I$. If f is n-times differentiable at x_0 $\forall n \in \mathbb{N}$, then we call

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

the <u>Taylor series</u> of f centered at x_0.

<u>Ex</u> $f : \mathbb{R} \to \mathbb{R}$, $f(x) = e^x$, $x_0 = 0$.

$$f^{(n)}(x) = e^x \implies f^{(n)}(0) = 1 \quad \forall n \geq 0$$

So the Taylor series of f centered at $x_0 = 0$ is

$$\sum_{n=0}^{\infty} \frac{1}{n!} x^n .$$

Radius of convergence:

$$c_n = \frac{1}{n!} \implies |c_n|^{\frac{1}{n}} = \frac{1}{\sqrt[n]{n!}} \to 0 \quad \text{(by lecture 15)}$$

$$\implies \limsup_{n \to \infty} |c_n|^{\frac{1}{n}} = 0 \implies R = +\infty.$$

So, by lecture 35, the series converges $\forall x \in (-\infty, \infty)$ and $g(x) = \sum \frac{1}{n!} x^n$ is continuous.

<u>Q</u>: In general,

1. Is the function g that we get the same as the function f that we started with?
2. When are the partial sums good approximations of f?

<u>A</u>: 1. Not always! (See HA13.)

2. Taylor's theorem:

<u>Thm</u> (Taylor's theorem with remainder) Let $I \subseteq \mathbb{R}$

be an open interval, $f: I \to \mathbb{R}$, and $x_0 \in I$.
If the nth derivative $f^{(n)}: I \to \mathbb{R}$ of f exists, then $\forall x \in I \setminus \{x_0\}$ $\exists y$ between x_0 and x s.t.

$$f(x) = \underbrace{\sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k}_{\text{(n-1)st Taylor polynomial approximation of } f} + \underbrace{\frac{f^{(n)}(y)}{n!}(x-x_0)^n}_{\text{remainder}}.$$

<u>Cor 1</u> Moreover, if also $f^{(n)}: I \to \mathbb{R}$ is bounded, then $\exists C_n > 0$ s.t.

$$\underbrace{\left| f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k \right|}_{\text{the error in the approximation}} \leq \frac{C_n}{n!} \underbrace{|x-x_0|^n}_{\text{is higher order}} \quad \forall x \in I.$$

<u>Pf</u>: Fix $x \in I$. By the theorem, $\exists y \in I$ s.t.

$$\left| f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k \right| = \frac{|f^{(n)}(y)|}{n!} |x-x_0|^n.$$

Pick $C_n > 0$ s.t. $|f^{(n)}(y)| \leq C_n \ \forall y \in I$. $\square$

<u>Cor 2</u> If all of the derivatives $f^{(n)}: I \to \mathbb{R}$ exist and $\exists C > 0$ s.t. $|f^{(n)}(x)| \leq C \ \forall x \in I \ \forall n \geq 0$, then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n \quad \forall x \in I.$$

- I.e., the Taylor series converges to a function g, and $g = f$.

Pf: Fix $x \in I$. By the theorem, for each $n \geq 1$ we have

$$\left| f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k \right|$$

$$= \frac{|f^{(n)}(y)|}{n!} |x-x_0|^n \quad \text{for some } y \in I$$

$$\leq \frac{C}{n!} |x-x_0|^n \to 0 \quad \text{as } n \to \infty.$$

(Note that $\forall r \in \mathbb{R}$, $\frac{r^n}{n!} \to 0$ as $n \to \infty$, since $\sum \frac{r^n}{n!}$ converges by the ratio test.)

So the series $\sum \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$ converges to $f(x)$. $\square$

Ex $f(x) = e^x$.

Claim: $\sum_{n=0}^{\infty} \frac{1}{n!} x^n = e^x \quad \forall x \in \mathbb{R}$. Fix $x_0 \in \mathbb{R}$.

Set $R = |x_0| + 1$. Then $f^{(n)} : (-R, R) \to \mathbb{R}$, $f^{(n)}(x) = e^x$ is bounded by e^R $\forall n$. So, by Corollary 2,

$$\sum_{n=0}^{\infty} \frac{1}{n!} x^n = e^x \quad \forall x \in (-R, R).$$

In particular, equality holds for $x = x_0$.